

Elizaveta Rebrova

Contact information

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 University of Michigan
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Research interests

Non-asymptotic random matrix theory
High-dimensional probability
Geometric functional analysis
Applications to high dimensional data analysis

Education

2013-present **University of Michigan - Ann Arbor**
 PhD candidate, Mathematics
 Advisor *Roman Vershynin*
 Expected graduation date: June 2018

2007-2012 **Lomonosov Moscow State University**
 Specialist degree in Mathematics (B.S.+M.S. equivalent)
 Diploma with honors
 Master thesis *Sobolev classes on infinite dimensional spaces*
 Advisor *Vladimir I. Bogachev*

2011-2012 **Yandex School of data analysis, Moscow**
 Computer Science Department

 12 Masters-level courses in computer science and data analysis, including
 machine learning, algorithms and data structures, computational linguistics,
 intro to R, game theory, applied statistics

Publications and preprints

2017 *Norms of random matrices: local and global problems* (with R. Vershynin).
 Advances in Mathematics, to appear. arXiv:1608.06953

2016 *Coverings of random ellipsoids, and invertibility of matrices with i.i.d.
heavy-tailed entries* (with K. Tikhomirov). Israel J. Math., to appear.
arXiv:1508.06690

2013 *Functions of bounded variation on infinite-dimensional spaces with measures*
 (with V. Bogachev). Doklady Mathematics (Doklady Akademii Nauk), 87(2)

Internship experience

2017 *Lawrence Berkeley National Lab, Berkeley, CA*

Scalable Solvers Group; internship mentor: Xiaoye Sherry Li
 Investigated applications of STRUMPACK linear solver to kernel matrices

Talks

- June, 2017 *Non-asymptotic spectral properties of heavy-tailed random matrices*, two talks, Universite-Paris-Est Marne-la-Vallee and Universit Paris Diderot, Paris, France
- May, 2017 *Norms of heavy-tailed random matrices: local and global problems*, University of Alberta, Edmonton, Canada
- March, 2017 *Local and global obstructions for the random matrix norm regularization*, MIC seminar at Center for Data Science, NYU, New York
- June, 2016 *Regularization of the norm of matrices with i.i.d. heavy-tailed entries*, CMS Summer Sectional Meeting University of Alberta, Edmonton, Canada
- March, 2016 *Coverings of random ellipsoids, and invertibility of matrices with i.i.d. heavy-tailed entries*, AMS Sectional Meeting, University of Georgia, Athens, GA

Seminar presentations at University of Michigan

- January, 2017 *Structured Random Matrices tutorial*, Analysis/Probability learning seminar, University of Michigan
- September, 2016 *Matrix regularizing effects of Gaussian perturbations*, Analysis/Probability learning seminar, University of Michigan
- October, 2015 *Geometric random walks*, Student Analysis, University of Michigan
- October, 2015 *Refined epsilon-nets and invertibility of random square matrices with i.i.d. heavy-tailed entries*, Analysis/Probability learning seminar, University of Michigan
- March, 2015 *The smallest singular value of random rectangular matrices with no moment assumptions on entries*, Analysis/Probability learning seminar, University of Michigan
- March, 2015 *Small ball probabilities and an introduction to the Littlewood-Offord problem*, Student Analysis seminar, University of Michigan
- November, 2014 *Bounds for the volume of the sections of the unit cube in R^n (by Keith Ball)*, Student Analysis seminar, University of Michigan
- 2014-2015 Series of talks on Geometric analysis reading seminar (topics included: contact points and John's theorem; Milman's quotient of subspace theorem and Bourgain-Milman inequality; Paouris deviation inequality, Banach-Mazur distance to the cube estimates), University of Michigan

Schools and workshops participation

- November, 2017 MSRI program workshop: Geometric functional analysis and applications, Berkeley, CA
- August, 2017 Introductory Workshop: phenomena in high dimensions, MSRI, Berkeley, CA
- August, 2017 Connections for Women: geometry and probability in high dimensions, MSRI, Berkeley, CA
- July, 2016 The Mathematics of Data, Graduate summer school, IAS/Park City Mathematics Institute, Utah
- June, 2016 CMS Summer Meeting, University of Edmonton, Alberta, Canada
- June, 2016 Summer School on Random Matrices, University of Michigan, Ann Arbor, MI

April, 2016	Informal Analysis Seminar, Kent State University
March, 2016	AMS Sectional Meeting, University of Georgia, Athens, GA
November, 2015	Informal Analysis Seminar, Kent State University
November, 2015	Analytic and Probabilistic Techniques in Modern Convex Geometry, University of Missouri
April, 2015	Workshop on information Theory and Concentration Phenomena, Institute for Mathematics and its Applications, Minneapolis
March, 2015	Informal Analysis Seminar, Kent State University
October, 2014	Informal Analysis Seminar, University of Michigan, Ann Arbor
May, 2014	Random Matrix Theory school, program for Women and Mathematics, IAS, Princeton
May, 2012	The XIX International student, postgraduate and young scientist conference “Lomonosov”, Moscow, Russia
July, 2006-2009	Summer school “Contemporary mathematics”, Dubna, Russia

Teaching and organization

2014-2017	Graduate Student Instructor, University of Michigan, Ann Arbor Calculus I (Fall 2014) Differential Equations (Winter 2015, Winter 2016, Fall 2016) Computational Finance (Fall 2017)
2013-2014	Instructor in Math Lab, University of Michigan, Ann Arbor
2012-2013	Algebra instructor, Kolmogorov math and physics high school, Moscow, Russia
2009-2012	Seminar work with math high school students, Moscow State School N 57, Moscow, Russia.
2007-2012	Member of Organizing Committee, Moscow Mathematical and Linguistics Olympiads, Moscow, Russia.

Languages

Computer	Python, Matlab, LaTeX
Human	English (fluent), Russian (native), German (basic), French (basic)

Fellowships, grants, awards

2017	NSF scholarship for summer research internship at Lawrence Berkeley National Laboratory
2017	Rackham One-Term Dissertation Fellowship, University of Michigan
2014-2016	Mathematics Department Graduate Summer Fellowship, University of Michigan
2012	RFFI 10-01-00518, research grant, Russia
2010	Excellent progress mayor’s scholarship, Moscow, Russia
2007	The 5th International Linguistics Olympiad (for high school students), 1st place team award and individual Bronze award